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THE EFFECTS OF COGNITIVE RESOURCE REQUIREMENTS, AVAILABILITY, AND ARGUMENT QUALITY ON BRAND ATTITUDES

A Melding of Elaboration Likelihood and Cognitive Resource Matching Theories

Keith S. Coulter and Girish N. Punj

ABSTRACT: Researchers have investigated a number of different psychological constructs that occur as a result of advertising exposure, constructs that contribute to the formation of attitudes toward the ad (A_{ad}) and brand (A_b). The Dual Mediation Model (DMM) (MacKenzie and Lutz 1989), which defines hypothesized interrelationships among these constructs, has been well supported in previous research. Other researchers have provided evidence that brand attitudes formed as a result of advertising exposure are greater if there is a "match" between the cognitive resources required to process the advertising message and those made available for processing (Anand and Sternthal 1989). The authors suggest that DMM constructs, as well as the links among those constructs, are moderated by the match between cognitive resource requirements and availability. Furthermore, the authors suggest that the nature of these moderating effects will depend on argument quality. The results of empirical testing of a modified version of the DMM under alternate processing-resource/argument-quality conditions are consistent with this theoretical explanation.

An ongoing concern of advertising practitioners is the persuasive impact of their marketing communications. One of the most widely employed measures of persuasive impact is change in brand attitude (A_b). Consequently, research attention continues to focus on describing the specific types of cognitive and affective responses to advertising that both lead to and result from brand attitude formation. The five types of responses most often studied in the ad effects literature are (1) affective responses to the ad, (2) ad cognitions, (3) attitude toward the ad, (4) brand cognitions, and (5) purchase intention. The latter four constructs, as well as brand attitude, are incorporated in the traditional Dual Mediation Model (DMM), which has been well supported within the ad effects literature (e.g., Brown and Stayman 1992; Homer 1990; MacKenzie and Spreng 1992).

The DMM is theoretically grounded in Petty, Cacioppo, and Schumann's (1983) notion of "elaboration likelihood" (i.e., the likelihood of generating thoughts in response to a particular stimulus). The degree of elaboration determines the relative strength of DMM paths, that is, which linkages are likely to predominate in any given stimulus exposure situation. When elaboration likelihood is high, viewers of a per-

suasive communication are prone to generate a relatively large number of ad- and brand-related thoughts, and to engage in more effortful (i.e., "central") processing of the ad's message content. Conversely, when elaboration likelihood is low, few resources are made available for message processing, and brand attitudes are more likely to be formed by means of heuristics (Eagly and Chaiken 1993), affect transfer (Lutz 1985), or other less effortful (i.e., "peripheral") message processing. With its inclusion of a linkage between attitude toward the ad and brand cognition, the DMM extended early elaboration likelihood theory to include the possibility that a peripheral cue (A_{ad}) could also have an impact on the central route to persuasion by fostering message acceptance (MacKenzie, Lutz, and Belch 1986). Thus, the DMM was instrumental in demonstrating how central and peripheral processes are interrelated.

The Elaboration Likelihood Model (ELM) predicts that motivation and ability influence the likelihood of message elaboration, and that increased elaboration enhances persuasion when the message is strong (i.e., primarily evokes support arguments) and diminishes persuasion when the message is weak (i.e., primarily evokes counterarguments) (Cacioppo and Petty 1979; Ito 2002; Petty, Cacioppo, and Goldman 1981; Petty and Wegener 1999). An argument is judged to be strong if it elicits the positive brand cognitions desired by the message communicator (Petty and Cacioppo 1986; Petty, Haughtvedt, and Smith 1995). Thus, a "weak" message typically cannot elicit more positive than negative brand-related cognitions, nor can a "strong" message result in a preponderance of negative (versus positive) brand cognitions.

The notion that argument quality is the primary driver of

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message acceptance runs counter to the results of a number of studies that have linked message discounting to cognitive resource allocation rather than argument quality (or strength) (e.g., Anand and Sternthal 1989, 1990; Edell and Staelin 1983; Gilbert, Tafarodi, and Malone 1993; Meyers-Levy and Peracchio 1995; Peracchio and Meyers-Levy 1997; Petty, Wells, and Brock 1976; Shiv, Edell, and Payne 1997). According to these studies, the impact of advertisements on brand evaluations may be sensitive to the relation between the cognitive resources required (RR) to process the message, and those made available (RA) for processing. More specifically, the cognitive resource matching (CRM) hypothesis predicts that any message (strong or weak) will enhance persuasion if there is a "match" between required and available cognitive resources (i.e., $RA = RR$) (Keller and Block 1997). Thus, a persuasive communication that results in a disproportionate number of positive to negative brand cognitions under low resource-allocation conditions could theoretically result in a disproportionate number of negative to positive brand cognitions under high resource-allocation conditions (or vice versa), depending on the processing requirements of the advertising stimuli.

In this paper, we refine the ELM framework by demonstrating that increased elaboration will enhance persuasion when the message is strong and diminish persuasion when the message is weak, but only within a certain range of processing motivation. That range is determined by the relation between required (RR) and available (RA) cognitive resources. In the case of a strong message, we demonstrate that increasing processing motivation (and hence elaboration) up to the level where $RA = RR$ will enhance persuasion; beyond that level, increased elaboration will diminish persuasion due to a greater amount of message discounting. In the case of a weak message, we demonstrate that increasing processing motivation up to the level where $RA = RR$ will diminish persuasion; beyond that level, no further diminution is found. Thus, we also refine the CRM framework by showing that persuasion will be enhanced if there is a match between required and available cognitive resources, but only in the case of a strong message. In demonstrating the conditions under which ELM and CRM theories digress and converge, we shed light on the manner in which processing motivation, resource requirements, and argument quality interact in determining brand attitudes.

THEORETICAL DEVELOPMENT

To successfully integrate dual-process (DMM) and resource allocation (CRM) theories, we must first respecify the DMM framework. As mentioned earlier, the traditional DMM specification includes five constructs (i.e., attitude toward the ad [A_{ad}], attitude toward the brand [A_b], ad cognitions [C_{ad}], brand cognitions, and purchase intention). Because cognitive responses to advertising can be either positive or negative,

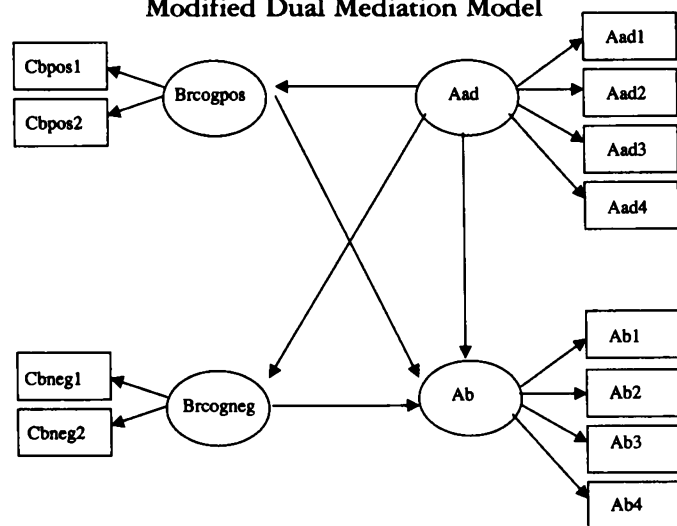
and because these constructs may act independently in terms of their influence on brand attitudes, we refine the brand cognitions construct to include both positive (i.e., $Brcog_{pos}$) and negative (i.e., $Brcog_{neg}$) components (MacKenzie and Spreng 1992). In addition, we exclude both purchase intention and ad cognitions from the model. The former is excluded because it is a consequent rather than an antecedent of A_b , and the primary concern of this paper is with the processes that impact persuasion and attitude change. The latter is excluded because its indirect effects on A_b are entirely mediated by (and thus captured within) the A_{ad} latent construct.¹ Our modified version of the DMM is depicted in Figure 1, and is used as the base model in our study.

As mentioned previously, studies have shown that brand attitudes are more positive when there is a match between required and available cognitive resources (Meyers-Levy and Peracchio 1995). The theory underlying these results is that most messages, particularly those in the form of commercial advertisements, are designed to be persuasive, and thus include elements that support the advocacy in a compelling manner (i.e., the assumption is one of a strong-message argument). Apportioning an amount of cognitive resources that just satisfies the level necessary for adequate comprehension and analysis of the message, then, should yield favorable results due to the generation of a relatively greater number of positive brand cognitions.

If the communication recipient allocates fewer resources to processing than the message requires ($RA < RR$), CRM theory predicts that persuasion is likely to be diminished due to incomplete, superficial, or inefficient message processing (Meyers-Levy and Malaviya 1999). In this case, we expect that fewer (i.e., fewer than "necessary") positive brand-related cognitions are generated. Alternatively, if the resources a person consigns to processing a persuasive communication surpass the level that the message effectively requires ($RA > RR$), the message recipient may generate additional advocacy-consonant cognitions. However, the recipient is more likely to commit excess resources to generating thoughts that either question the message assertions (i.e., counterarguments) or are advocacy-irrelevant (idiosyncratic) in nature (Keller and Block 1997).

Researchers have suggested that the preponderance of negative cognitions that occur under $RA > RR$ conditions may be the result of boredom, tedium, or "wear-out" effects (Anand and Sternthal 1989, 1990; Edell and Staelin 1983; Gilbert, Tafarodi, and Malone 1993; MacInnis and Jaworski 1989). Other research seems to suggest (building on Friestad and Wright's [1994] Persuasion Knowledge Model) that participants may begin to anticipate manipulative intent in commercial messages (and hence generate more negative brand cognitions) when the allocation of a high level of resources reduces cognitive capacity (Campbell and Kirmani 2000).

FIGURE 1
Modified Dual Mediation Model



Alternatively, if increasing the level of available resources up to the point where $RA > RR$ complicates the decision-making process, then research has shown that the decision task may be simplified by placing greater weight on negative information about the subject of that task (Wright 1974; Wright and Weitz 1977). This would also likely result in more negative thoughts.

The authors speculate that the preponderance of negative cognitions that occurs under $RA > RR$ conditions may stem from the fact that an effective evaluation involves assessing both the positives and negatives of any particular stimulus. If the positives are readily apparent (as would be the case with a strong-message argument), then the “easiest” form of evaluation (requiring the least cognitive resources) would involve assessing and reporting those positive aspects. According to the “sufficiency principle” of Chaiken, Liberman, and Eagly (1989), if actual confidence in that evaluation is undermined as the evaluator becomes motivated (perhaps due to an external source) toward further scrutiny, he or she may begin looking for the less apparent faults (negatives) that may have been missed on initial inspection (Eagly and Chaiken 1993). Exposure of those faults would involve the generation of negative brand cognitions.

In terms of model paths, we would argue that if the message recipient allocates a level of resources that just meets the level required for adequate comprehension and scrutiny of the message ($RA = RR$), then attendance to that message is the *only* cognitive function attended to, and the salience of message-intended positive thoughts should be high. The increased salience of the positive brand cognitions should cause the intended message to be processed more effectively (Anand and Sternthal 1989), leading to a stronger $Brcog_{pos} - A_b$ relation, and thus more favorable judgments.

In sum, we expect that increasing available resources up to the level where $RA = RR$ should enhance persuasion when

the message is strong, due to both a relatively greater increase in the number of positive versus negative brand cognitions, and a stronger $Brcog_{pos} - A_b$ relation. However, if available resources are increased beyond the $RA = RR$ level, such that $RA > RR$, then persuasion should be diminished due to a relatively greater increase in the number of negative brand cognitions. Thus, the following hypotheses are proposed:

H1: For an ad with a strong-message argument, (a) the mean level of $Brcog_{pos}$ will be greater, (b) the $Brcog_{pos} - A_b$ relation will be stronger, and (c) mean A_b will be greater under $RA = RR$ than under $RA < RR$ conditions.

H2: For an ad with a strong-message argument, (a) the mean level of $Brcog_{neg}$ will be greater, and (b) mean A_b will be lower under $RA > RR$ than under $RA = RR$ conditions.

As noted earlier, ELM theory predicts that increased elaboration will diminish persuasion when a message is weak (Friedrich and Smith 1998). Assuming a weak message is defined as one that has the same number of but less persuasive arguments than a strong message (Petty, Cacioppo, and Schumann 1983), the cognitive resources required to process the weak message should be the same as those required to process the strong message. However, apportioning an amount of cognitive resources that just satisfies the level necessary for adequate comprehension and analysis of the message (i.e., $RA = RR$) should not yield more favorable brand evaluations relative to the $RA < RR$ or $RA > RR$ conditions, because effective and accurate comprehension would enable the message recipient to deduce that the message is *not* particularly strong. In effect, the recipient is better able to assess the inadequacies of the message claims. Thus, as available resources are increased in going from the $RA < RR$ to $RA = RR$ condition, we expect a greater relative increase in the number of negative, rather than positive, brand cognitions. Furthermore, we anticipate that the $Brcog_{neg} - A_b$ path should also increase in strength. Under weak-message argument conditions, thoughts may be directed toward elements of the message other than those intended by the message communicator. A focus on unintended message elements should lead to the reduced salience of any positive brand cognitions associated with that message. In addition, the expected increase in number of negative brand cognitions in the $RA = RR$ condition should increase the salience of counterarguments relative to the positive message elements that the persuasive communication was meant to convey. The increased salience of negative brand cognitions, in turn, should increase their importance in forming brand evaluations, leading to a stronger $Brcog_{neg} - A_b$ path.

Finally, as mentioned earlier, both tedium (e.g., Anand and Sternthal 1989, 1990) and undermined confidence (e.g., Chaiken, Liberman, and Eagly 1989) have been proposed as

possible explanations for the relatively greater number of $Brcog_{neg}$ under $RA > RR$ conditions. Both explanations assume a greater proportion of $Brcog_{pos}$ when available cognitive resources match cognitive resource requirements. But if the $RA = RR$ condition is instead characterized by a preponderance of *negative* brand cognitions, then (assuming either of these explanations is correct) a further increase in available resources should engender the curtailment of negative, rather than positive, brand-related thought. As negative thought is curtailed, the message recipient should focus more on the positive aspects of the message, leading to an increase in the number of $Brcog_{pos}$. In addition, there should be no further increase in the salience of those negative cognitions that do occur. Thus, if available resources are increased beyond the $RA = RR$ level, persuasion should be enhanced. We hypothesize that

H3: For an ad with a weak-message argument, (a) the mean level of $Brcog_{neg}$ will be greater, (b) the $Brcog_{neg} - A_b$ relation will be stronger, and (c) mean A_b will be lower under $RA = RR$ than under $RA < RR$ conditions.

H4: For an ad with a weak-message argument, (a) the mean level of $Brcog_{pos}$ will be greater, and (b) mean A_b will be greater under $RA > RR$ than under $RA = RR$ conditions.

All hypotheses are graphically represented in Table 1.

RESEARCH DESIGN

Processing Motivation

Consistent with prior research (Peracchio and Meyers-Levy 1997), we manipulate the level of resource availability (RA) by varying processing motivation. Processing motivation is manipulated by varying the initial experimental instructions (Coulter and Punj 1999; Homer 1990; MacKenzie and Lutz 1989). In our low processing-motivation (low RA) condition, the instructions encouraged participants to relax and examine the print ad as if they were sitting "in the comfort of their own living room." The instructions in our moderate processing-motivation (moderate RA) condition encouraged participants to examine the ad as if they were "in the market" for the advertised product, and the instructions in our high processing-motivation (high RA) condition encouraged participants to examine the ad as if they were in the market for the advertised product because they would later be asked to choose among brands.

Advertising Stimuli

To manipulate argument strength and resource requirements (RR), a print ad was constructed for a fictitious brand of clothing: "Lois Jeans." A female model appeared in the ad wearing the aforementioned product, with the headline reading "Lois

Jeans, Built to Fit You Right," and the message copy dealing with different physical attributes of the jeans. Several different variations of the ad were developed using different numbers of physical attributes. Pretesting ($n = 48$) revealed that exposure to an eight-attribute version resulted in a significantly ($p < .05$) greater number of cognitions than exposure to a four-attribute version, and a significantly ($p < .05$) smaller number of cognitions than exposure to a twelve-attribute version. Furthermore, the number of cognitions associated with the twelve-attribute version was not significantly different from the number associated with a sixteen-attribute version.² Thus, the version containing eight attributes was presumed to require "moderate" resource requirements.

Two different moderate resource requirement ads were then created for use in the study. The ad with the strong-message argument listed eight positive attributes of the jeans that could be evaluated by viewing the model in the illustration (e.g., stylish, great fit; see Appendix). The ad with the weak-message argument was identical to the ad with the strong-message argument; however, the eight attributes were either less discernable (e.g., stain resistant) or of questionable value (e.g., moderate shrinkage). The RA/RR conditions associated with each of our processing-motivation/argument-quality treatments are summarized in Table 1.

Participants and Procedure

A total of 361 female undergraduate liberal arts students from a major U.S. university took part in our study. Study participants received extra course credit. Females were chosen because the product was specifically geared toward this target segment. Upon arrival at the experimental site, participants were randomly assigned to one of the six processing-motivation/argument-strength conditions. After receiving initial instructions and viewing the target ads, participants filled out a written questionnaire designed to assess their reactions to those ads.

DATA AND MEASUREMENT

Independent and Dependent Variables

The two indicators used to measure the $Brcog_{pos}$ and $Brcog_{neg}$ structural variables were derived from a verbal protocol in which participants were asked to write down all thoughts that occurred to them during exposure to the advertisement. Subsequently, two judges who were unfamiliar with the study objectives independently coded these protocols as either (1) positive (Cb_{pos1}) or negative (Cb_{neg1}) brand cognitive statements related to overall brand impressions, (2) positive (Cb_{pos2}) or negative (Cb_{neg2}) brand cognitive statements related to specific message content, or (3) idiosyncratic (non-ad- or non-brand-related) thoughts. "Proportional reduction in loss" measures of

TABLE I
Graphical Representations of Hypotheses and Resource Requirement/Availability Manipulations

		Low RA (PM) (RA < RR)	Moderate RA (PM) (RA = RR)	High RA (PM) (RA > RR)
H1–H2 (eight strong arguments, moderate RR, strong message)	$Brcog_{pos}$	++	++++	++++
	$Brcog_{neg}$	—	—	—
	$Brcog_{pos}-A_b$	++	+++	+++
	$Brcog_{neg}-A_b$	*	*	*
	A_b	+	++	+
H3–H4 (eight weak arguments, moderate RR, weak message)	$Brcog_{pos}$	+	++	++++
	$Brcog_{neg}$	—	—	—
	$Brcog_{pos}-A_b$	*	*	*
	$Brcog_{neg}-A_b$	—	—	—
	A_b	—	—	—

Notes: RR = resource requirements; RA = resource availability; PM = processing motivation.

* Inferences are not made regarding these relations.

intercoder reliability (Rust and Cooil 1994) were .84, .86, .77, and .82 for the four brand-related cognitive categories, respectively, and .92 for the idiosyncratic thought category. For the final classification, disagreement between the judges was resolved by discussion until a consensus was reached.

Both ad and brand attitudes were measured by four, seven-point semantic differential items: like/dislike, good/bad, positive/negative, and favorable/unfavorable (Batra and Ray 1986; Edell and Burke 1987; MacKenzie, Lutz, and Belch 1986; Mitchell and Olson 1981). The scales formed by the unweighted sums of the four items used to form the ad and brand attitude scales had Cronbach α 's of .88 and .92, respectively.

Manipulation Checks

A measure of attention was used as a processing motivation manipulation check (Laczniak and Muehling 1993). The measure was formed by averaging across two, seven-point Likert scale items (paid attention to the message in the ad; did not pay attention to the message communicated). Successful experimental manipulation would be indicated if the high processing-motivation groups paid more attention than the moderate processing-motivation groups, and if the moderate processing-motivation groups paid more attention than the low processing-motivation groups. Because level of processing motivation determines resource availability, we expected that the mean number of total brand cognitions would be greater under high than under moderate or low processing-motivation conditions.

Participants indicated how strong the overall message arguments were on two, seven-point semantic differential scale items

anchored by believable/not believable and convincing/not convincing (Petty, Cacioppo, and Schumann 1983). Argument strength would be successfully manipulated if ad claims were significantly more believable and convincing for participants exposed to the strong-message ad than for participants exposed to the weak-message ad. In addition, participants indicated how difficult ad-claim processing was using two, seven-point semantic differential scale items anchored by easy/difficult and simple/complicated (Meyers-Levy and Peracchio 1995). Resource requirements would be successfully manipulated if ad-claim processing was not significantly different across ads.

RESULTS

Manipulation Checks

ANOVA (analysis of variance) results and subsequent Scheffé ($p \leq .05$) comparisons revealed that participants paid significantly more attention to the message, $F(2, 352) = 10.19$, $p < .001$, in the high processing-motivation conditions ($\bar{x} = 4.92$) than in the moderate processing-motivation conditions ($\bar{x} = 3.28$), and in the moderate than in the low processing-motivation conditions ($\bar{x} = 2.01$). There was no argument-strength main effect, and no processing motivation $\times$ argument strength interaction. ANOVA results and subsequent Scheffé ($p \leq .05$) comparisons also revealed that mean number of total cognitions was significantly greater, $F(2, 353) = 9.37$, $p \leq .01$, in the high processing-motivation conditions ($\bar{x} = 6.59$) than in the moderate processing-motivation conditions ($\bar{x} = 3.98$), and in the moderate than in the low processing-motivation conditions ($\bar{x} = 2.41$). There was no other main effect or interaction. Thus,

processing motivation (resource availability) appears to have been successfully manipulated.

ANOVA results and subsequent Scheffé ($p \leq .05$) comparisons revealed that ad message claims were significantly more believable, $F(1, 350) = 5.07, p \leq .05$, and convincing, $F(1, 349) = 4.82, p \leq .05$, for participants exposed to the strong-message ad ($\bar{x} = 3.81$, believable; $\bar{x} = 3.91$, convincing) than for participants exposed to the weak-message ad ($\bar{x} = 2.74$, believable; $\bar{x} = 2.57$, convincing). There was no processing-motivation main effect, and no argument strength $\times$ processing motivation interaction. Thus, argument strength appears to have been successfully manipulated.

ANOVA results and subsequent Scheffé ($p \leq .05$) comparisons also revealed no significant differences in ad-claim processing across strong ($\bar{x} = 3.91$, difficult; $\bar{x} = 4.03$, complicated) versus weak ($\bar{x} = 3.83$, difficult; $\bar{x} = 3.79$, complicated) advertisements. Thus, the resource requirements of our advertising stimuli were essentially equivalent. A comparison of A_{ad} means across groups revealed a significantly greater, $F(1, 350) = 5.88, p \leq .05$, A_{ad} mean for the strong-message ad ($\bar{x} = 4.96$) than for the weak-message ad ($\bar{x} = 2.35$), but no significant differences ($p > .05$) across processing motivation groups and no processing motivation $\times$ ad type interaction. Thus, A_{ad} was not expected to account for hypothesis test results.

Examination of the Base Model

To explain the variation in model paths and construct means/intercepts, we employed AMOS 4.0 (Arbuckle 1999) for all structural equation modeling (SEM) analysis. Consistent with prior research in this area (e.g., MacKenzie and Spreng 1992), each of the four ad- and brand-related items used to create the aforementioned ad- and brand-attitude scales was used as a separate A_{ad} or A_b indicator. To set the metric and ensure model identification, we constrained the value of the first indicator loading for each latent construct equal to 1 (Kenny 1979; Steenkamp and Baumgartner 1998).

Confirmatory factor analysis (CFA) involving the measurement model revealed that all indicator loadings were significant ($p \leq .05$) and within an acceptable range (.58–.94). The CFA provided a good fit to the data, $\chi^2(48) = 46.56, p = .432$, RMSEA (root mean square error of approximation) = .03, CFI (comparative fit index) = .98, and no modification indices of substantive magnitude emerged for the λ matrices, implying acceptable fit of the measurement model.

Next, we simultaneously estimated the measurement and structural models with participants pooled across treatment groups. Once again, our overall ("across groups") model provided a relatively good fit to the data, $\chi^2(53) = 122.15, p = .382$, RMSEA = .04, GFI (goodness-of-fit index) = .96, CFI = .97, demonstrating that the model is statistically plausible and can reasonably reproduce the correlation matrix

(Bagozzi and Yi 1988) (see Table 2). All (nonstandardized) indicator loadings fell within an acceptable range (.26–1.23) and were statistically significant ($p \leq .05$) (see Table 3). Examination of the structural path parameter estimates revealed that all of the proposed path relations were significant (see Table 4).

A comparison of path strengths and latent construct means/intercepts across processing motivation $\times$ argument strength subsamples required that the measurement models be equivalent for each group. Measurement parameters for the six groups were compared by first estimating the base model simultaneously for each subsample, $\chi^2(318) = 443.42, p = .215$, and then estimating the same base model while constraining the (previously *unconstrained*) measurement parameters to be equal across groups, $\chi^2(358) = 503.54, p = .157$. The $\chi^2_{diff}(40) = 60.12$ was not significant ($p > .05$), indicating that the measurement structures were essentially equivalent.

To compare construct means or intercepts across groups, we also needed to test for the invariance of item intercepts (i.e., "scalar" invariance; Steenkamp and Baumgartner, 1998). Item intercepts for the six groups were compared by first estimating the base model simultaneously for each group with measurement parameters constrained, $\chi^2(358) = 503.54, p = .157$, and then estimating the same model while individually constraining first the Cb_{pos} item intercepts, followed by the Cb_{neg} and A_b item intercepts (i.e., one at a time across the six groups, for example: $Cb_{pos1}[\text{Group 1}] = Cb_{pos1}[\text{Group 2}] = Cb_{pos1}[\text{Group 3}]$, etc.). In each case, the $\chi^2_{diff}(df = 5)$ was nonsignificant, and the RMSEA, CAIC, NNFI (non-normed fit index), and TLI (Tucker-Lewis index) indices indicated no substantial deterioration in model fit (Byrne, Shavelson, and Muthen 1989; Meredith 1993). Thus, the latent means could be meaningfully compared across groups.

Hypothesis Tests

The multiple-group model with measurement structures constrained, $\chi^2(358) = 503.54, p = .157$, was used as the basis of comparison for all hypothesis tests (Kenny 1979; Steenkamp and Baumgartner 1998). This model was compared with a series of models in which additional specific parameter(s) were constrained. In each case, the parameter of interest was constrained to be equal across the two groups hypothesized to differ. Hypotheses were confirmed if (1) the additional constraint model demonstrated a significantly ($p \leq .05$) poorer fit, $\chi^2_{diff}(1) > 3.84$, than the original model, and (2) examination of the structural parameter estimates revealed that this difference was in the predicted direction.

Strong-Argument Ad

In terms of the strong-argument ad, analysis revealed that the mean level of $Brcog_{pos}$, $\chi^2_{diff}(1) = 11.24, p \leq .01$, the $Brcog_{pos}$ —

TABLE 2
Pooled Sample Correlation Matrix
(n = 361)

Indicator	A _{ad1}	A _{ad2}	A _{ad3}	A _{ad4}	A _{b1}	A _{b2}	A _{b3}	A _{b4}	Cb _{neg1}	Cb _{neg2}	Cb _{pos1}	Cb _{pos2}
A _{ad1}	1.00											
A _{ad2}	.53	1.00										
A _{ad3}	.54	.53	1.00									
A _{ad4}	.47	.40	.54	1.00								
A _{b1}	.19	.34	.20	.26	1.00							
A _{b2}	.24	.27	.24	.19	.36	1.00						
A _{b3}	.21	.19	.19	.22	.47	.41	1.00					
A _{b4}	.24	.27	.24	.24	.40	.37	.43	1.00				
Cb _{neg1}	-.07	-.16	-.18	-.09	-.12	-.15	-.15	-.18	1.00			
Cb _{neg2}	-.10	-.09	-.11	-.08	-.11	-.13	-.17	-.16	.53	1.00		
Cb _{pos1}	.27	.24	.28	.21	.38	.29	.35	.29	-.01	-.03	1.00	
Cb _{pos2}	.26	.26	.26	.10	.27	.27	.28	.25	-.03	-.01	.51	1.00

TABLE 3
Nonstandardized Parameter Estimates: Measurement Model Pooled Sample (n = 361)

Indicator	Mean	SE	CR	Loading	SE	CR
Cb _{pos1}	.91	.029	31.47	1.00	—	—
Cb _{pos2}	.93	.029	32.16	.26	.029	8.99
Cb _{neg1}	1.03	.025	40.96	1.00	—	—
Cb _{neg2}	1.11	.026	42.04	.28	.044	6.47
A _{ad1}	3.74	.104	36.05	1.00	—	—
A _{ad2}	3.48	.102	34.17	1.12	.098	11.4
A _{ad3}	3.85	.101	38.27	1.23	.094	13.0
A _{ad4}	3.68	.107	34.43	1.09	.105	10.5
A _{b1}	3.06	.090	33.98	1.00	—	—
A _{b2}	3.27	.091	36.07	.58	.078	7.5
A _{b3}	3.38	.086	39.46	.64	.077	8.3
A _{b4}	3.31	.094	35.05	.64	.082	7.8

Notes: CR = critical ratio (critical ratios > 2.00 are significant at $p \leq .05$); SE = standard error.

A_b relation, $\chi^2_{diff}(1) = 9.18, p \leq .01$, and mean brand attitude (A_b), $\chi^2_{diff}(1) = 7.15, p \leq .05$, were all greater under moderate (RA = RR) processing-motivation conditions than under low (RA < RR) processing-motivation conditions (see Tables 4 and 5). Thus, H1a, H1b, and H1c are supported. Analysis further revealed that the mean level of *Brcog_{neg}* was significantly greater, $\chi^2_{diff}(1) = 12.69, p \leq .01$, whereas mean A_b was significantly lower, $\chi^2_{diff}(1) = 5.26, p \leq .05$, under high (RA > RR) processing-motivation conditions than under moderate (RA = RR) processing-motivation conditions (see Tables 4 and 5). Hence, H2a and H2b are also supported.

Weak-Argument Ad

As expected, analysis revealed that the mean level of *Brcog_{neg}*, $\chi^2_{diff}(1) = 15.31, p \leq .01$, and the *Brcog_{neg}*-A_b relation, $\chi^2_{diff}(1) = 6.22, p \leq .05$, were greater, whereas mean A_b was lower, $\chi^2_{diff}(1) = 8.39, p \leq .01$, under moderate (RA = RR) processing-motivation conditions than under low (RA < RR) pro-

cessing-motivation conditions (see Tables 4 and 5). Thus, H3a, H3b, and H3c are supported. Analysis further revealed that the mean level of *Brcog_{pos}*, $\chi^2_{diff}(1) = 6.13, p \leq .05$, was significantly greater under high (RA > RR) processing-motivation conditions than under moderate (RA = RR) processing-motivation conditions, but that there was no significant difference in mean A_b, $\chi^2_{diff}(1) = 1.87, p > .05$ (see Tables 4 and 5). Thus, H4a is supported, but H4b is not supported.

SUMMARY AND DISCUSSION

Our results indicate that message discounting may be linked not only to the nature of the message arguments (i.e., strong versus weak), but also to the cognitive resources allocated by the message recipient. Increasing processing motivation (elaboration) enhanced persuasion when the message was strong and diminished persuasion when the message was weak, but only up to the point where available resources matched

TABLE 4
Nonstandardized Structural Parameter Estimates: Overall Pooled Sample and Individual Groups

Parameter	Number and strength of arguments and motivation level						
	Eight strong arguments				Eight weak arguments		
	Overall (n = 361)	Low (n = 60)	Moderate (n = 59)	High (n = 61)	Low (n = 58)	Moderate (n = 61)	High (n = 62)
A_{ad}	.29	.37	.34	.39	.22	.21	.23
SE	.08	.05	.12	.13	.09	.10	.08
A_b	.48	.70	.95	.40	.39	.25	.26
SE	.16	.15	.22	.14	.07	.12	.09
$Brcog_{pos}$	.98	.60	1.26	1.29	.51	1.03	1.20
SE	.08	.06	.11	.12	.08	.07	.04
$Brcog_{neg}$	1.12	.53	1.11	1.85	.76	1.44	1.39
SE	.10	.05	.12	.13	.06	.09	.07
$A_{ad} - Brcog_{pos}$	.50	.42	.57	.59	.40	.48	.51
SE	.09	.03	.04	.12	.11	.05	.13
$A_{ad} - Brcog_{neg}$	-.28	-.17	-.36	-.35	-.21	-.32	-.31
SE	.07	.04	.13	.09	.03	.08	.07
$Brcog_{pos} - A_b$	.30	.21	.53	.46	.23	.17	.18
SE	.11	.04	.07	.12	.05	.10	.09
$Brcog_{neg} - A_b$	.15	-.09	-.11	-.14	-.12	-.20	-.18
SE	.07	.02	.05	.03	.05	.09	.05
$A_{ad} - A_b$	.27	.39	.24	.20	.38	.25	.22
SE	.08	.11	.08	.09	.13	.10	.04

Notes: SE = standard error; all pooled sample path estimates are significant at $p \leq .05$.

TABLE 5
Hypotheses Test Results

Hypotheses	Parameter	Constraint	df	χ^2	χ^2_{diff}	p value
H1a	$Brcog_{pos}$	A = B	359	514.78	(1) = 11.24	$\leq .01$
H1b	$Brcog_{pos} - A_b$	A = B	359	512.72	(1) = 9.18	$\leq .01$
H1c	A_b	A = B	359	510.69	(1) = 7.15	$\leq .05$
H2a	$Brcog_{neg}$	B = C	359	516.23	(1) = 12.69	$\leq .01$
H2b	A_b	B = C	359	508.80	(1) = 5.26	$\leq .05$
H3a	$Brcog_{neg}$	D = E	359	518.85	(1) = 15.31	$\leq .01$
H3b	$Brcog_{neg} - A_b$	D = E	359	509.76	(1) = 6.22	$\leq .05$
H3c	A_b	D = E	359	511.93	(1) = 8.39	$\leq .01$
H4a	$Brcog_{pos}$	E = F	359	509.67	(1) = 6.13	$\leq .05$
H4b	A_b	E = F	359	505.41	(1) = 1.87	$> .05$

Note: The multiple-group model with measurement structures constrained, $\chi^2(358) = 503.54$, $p = .157$, is used as the initial basis of comparison for hypotheses tests.

- A. Low processing motivation, moderate resource requirement (strong) ad.
- B. Moderate processing motivation, moderate resource requirement (strong) ad.
- C. High processing motivation, moderate resource requirement (strong) ad.
- D. Low processing motivation, moderate resource requirement (weak) ad.
- E. Moderate processing motivation, moderate resource requirement (weak) ad.
- F. High processing motivation, moderate resource requirement (weak) ad.

resource requirements. The more positive brand attitudes associated with the strong message occurred as a result of both an increase in the mean level of positive brand cognitions and an increase in the strength of the $Brcog_{pos} - A_b$ relation. The less positive brand attitudes associated with the weak message occurred as a result of both an increase in the mean level of negative brand cognitions and an increase in the strength of the $Brcog_{neg} - A_b$ relation. Although both of these results are consistent with ELM theory (at least within the $RA < RR$ to $RA = RR$ range), the latter finding of less positive brand attitudes at moderate processing-motivation levels (where $RA = RR$) is inconsistent with CRM theory.

In the case of a strong message, increasing processing motivation (and hence resource availability) beyond the moderate level was found to diminish persuasion, due to a greater amount of message discounting. In the case of a weak message, increasing processing motivation beyond the moderate level (such that $RA > RR$) resulted in an increased amount of support argumentation, although no significant difference in brand attitude was observed. These results indicate that the strength of an argument is partially dependent on the circumstances under which the advertisement is viewed, and cannot be linked entirely to the task stimulus.

Our findings indicate that CRM theory may not be applicable (or at least robust) in the case of weak-message arguments. When $RA = RR$ under weak-message argument conditions, the consumer appears to be better able to assess or evaluate the inadequacies of the message claims. This leads to *less*, rather than more, favorable brand attitudes. In addition, as processing motivation is increased and available cognitive resources exceed cognitive resource requirements, consumers are led to increase their production of positive, rather than negative, brand cognitions. This seems to imply that wear-out, tedium (Anand and Sternthal 1989), or undermined confidence (Chaiken, Liberman, and Eagly 1989) rather than persuasion knowledge (Campbell and Kirmani 2000; Friestad and Wright 1994) or task complication (Wright 1974; Wright and Weitz 1977) are the primary drivers of the observed effects. A more pronounced anticipation of manipulative intent or a more complicated decision task would theoretically lead to a greater number of negative brand cognitions under $RA > RR$ conditions, regardless of argument quality.

In this study, we define a weak message as one that has the same number of arguments as a strong message, but the arguments are less persuasive (Petty, Cacioppo, and Schumann 1983). A weak message can also be defined as one that includes fewer (strong) brand assertions, rather than weaker message arguments (Petty and Cacioppo 1984). In the latter instance, argument quality is linked directly to resource requirements. If we had defined argument quality in terms of resource requirements, such that $RA = RR$ at a low level of processing motivation, and if we had found the same decrease

in brand attitude at moderate processing-motivation levels, then our findings would have been consistent with both CRM and ELM theory. The conclusion one must draw is that both resource requirements and argument quality are relative terms that must be examined in terms of the context in which they are presented. Future research efforts might be directed toward a more thorough examination of how these relative assessments impact brand attitudes.

Future research efforts might also be directed toward examining the effects of resource requirements and argument strength, when argument strength is manipulated through different combinations of positive and negative attributes. This is important because research has shown that weak arguments can "suppress" the impact of accompanying strong ones when considered in combination (Friedrich et al. 1996; Friedrich and Smith 1998). Furthermore, the weak arguments may reduce the impact of the accompanying strong arguments, either independently or because the strong arguments are evaluated in the "context" of the weak ones (Friedrich and Smith 1998). If the latter explanation is correct, then number of brand cognitions may not be directly related to processing resource requirements.

A related issue pertains to the absolute numbers of positive and negative brand cognitions generated in response to a persuasive communication, that is, to resource availability. As noted earlier, according to ELM theory, a strong-message argument is assumed to result in a greater number of positive than negative brand cognitions, resulting in more favorable brand evaluations (Petty and Wegener 1999). However, current theory does not postulate as to whether the numbers of positive and negative cognitions should be considered in absolute or relative terms (Wegener et al. 1995). For example, a participant viewing an ad could generate six positive brand cognitions and two negative brand cognitions, for a net of four positive brand cognitions. A second participant could generate four positive brand cognitions and one negative brand cognition, for a net of three positive brand cognitions. In absolute terms, the first participant has generated a greater number of favorable brand cognitions, and therefore should have a more favorable brand attitude (Petty and Wegener 1999). However, if one considers the proportion of positive to negative thoughts (Wegener et al. 1995), the first participant has generated three times more positive than negative thoughts, whereas the second participant has generated four times more positive than negative thoughts; therefore, brand attitude should be more favorable in the case of the second participant.

Our results show that absolute number of positive and negative brand cognitions changed within the $RA < RR$ to $RA = RR$ resource availability range, where ELM predictions appear most robust. Subsequent analysis using identical procedures revealed that the proportion of positive to negative brand cognitions did *not* change within that range. Thus, proportion

seems to be the more reliable indicator of argument quality.

The practical implications of our research are similar to those of extant cognitive resource matching theory. In essence, one should attempt to balance the cognitive resources that are required for message processing with those that are likely to be made available by the target audience, because attitude formation is likely to be optimal under such viewing conditions.

A possible limitation of our study involves the fact that we largely ignore the affective content of ad stimuli and the responses they elicit. Particularly in the resource-matching perspective, ad stimuli appear to differ along a single cognitive dimension. In the real world, however, persuasive communications may also require significant *affective* "resources" (e.g., empathy) to be fully processed. This may be a significant limitation of the CRM perspective. It is possible that responding affectively to ads in the manner sought by the advertiser also requires resources that are qualitatively different and arise out of a different cognitive set than resources that are focused on the processing of arguments. Thus, there may be a trade-off between the amounts of cognitive and affective resources required to process ads. We suggest that future research efforts be directed toward this important field of inquiry.

NOTES

1. We tested the effect of C_{ad} on A_b , controlling for A_{ad} , and found the effect to be nonsignificant. See Baron and Kenny (1986) for a comprehensive discussion of mediation testing procedures.

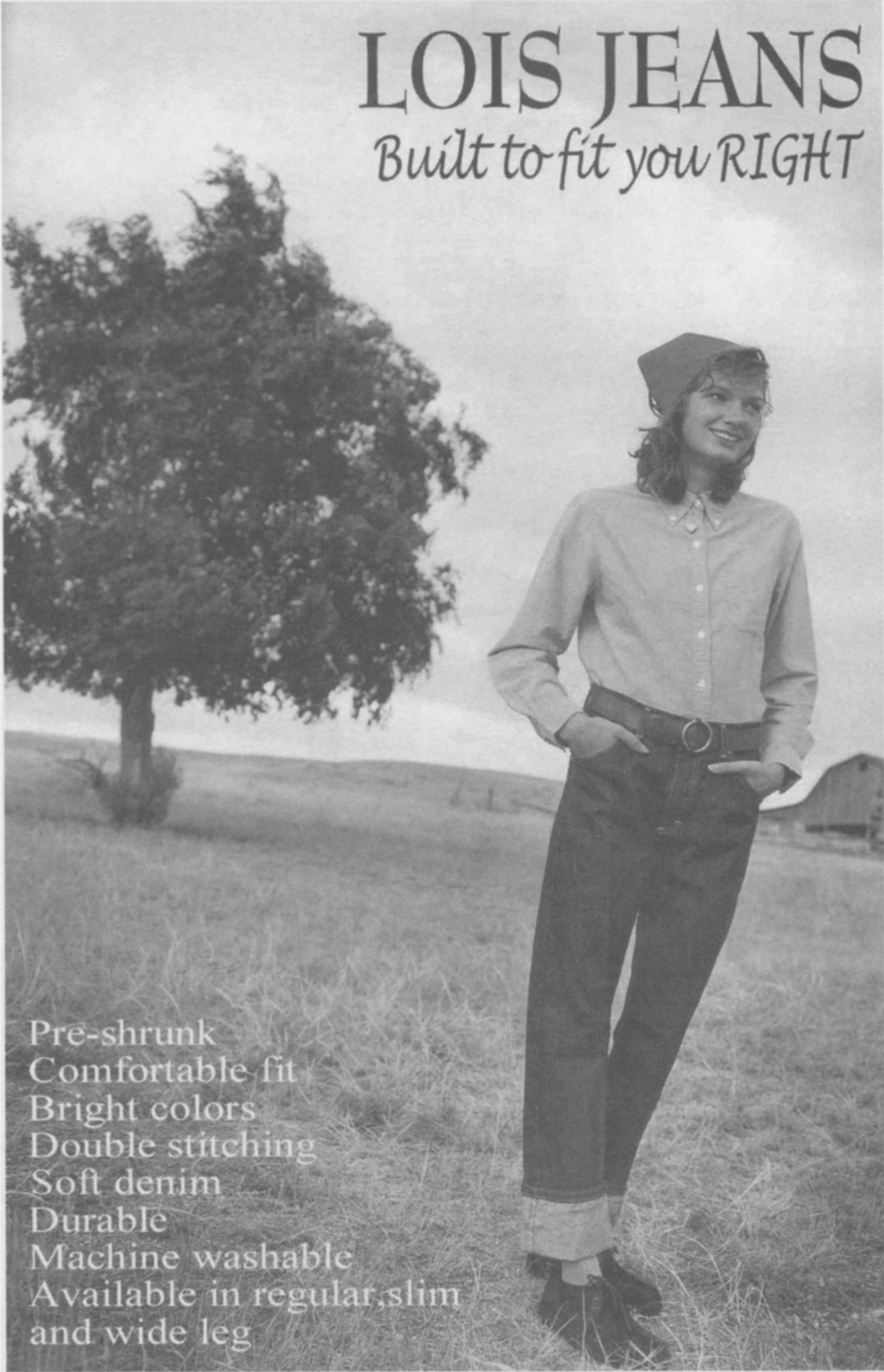
2. Pretesting also revealed that ad-claim processing was significantly ($p < .05$) more "difficult" and "complicated" for the 8-attribute version than for the 4-attribute version, and significantly less "difficult" and "complicated" for the 8-attribute version than for the 12-attribute version (see "Manipulation Check" section for description of measures).

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APPENDIX
Strong-Message Ad



LOIS JEANS
Built to fit you RIGHT

Pre-shrunk
Comfortable fit
Bright colors
Double stitching
Soft denim
Durable
Machine washable
Available in regular, slim
and wide leg

Note: Attributes for the weak-message ad were stain resistance, moderate shrinkage, embroidered logo, machine sewn, several colors, authentic denim, breathable fabric, and plenty of legroom.